

ANSWER KEY

SAT Practice Test 5 - Standardized Edition

Section 1: Reading and Writing

Module 1 | Module 2 Easier | Module 2 Harder

Section 2: Math

Module 1 | Module 2 Easier | Module 2 Harder

Answers and explanations are preserved from the original source file.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Q1. A) staggering

The passage emphasizes the "extraordinary range" and diversity of marine ecosystems. "Staggering" (meaning overwhelmingly great) best captures this idea of immense variety.

Q2. A) unfeasible

With over 1.2 million objects cataloged and each name needing to be unique, naming every one would be impractical or unfeasible. "Controversial" and "redundant" don't fit the logistical impossibility described.

Q3. C) resistant to

Commuters who already believe the fare is too expensive would naturally oppose (be resistant to) proposals to raise it further. The passage then says advocates should target a "more receptive group," confirming the first group is unreceptive.

Q4. A) inflicted by

The sentence describes damage caused by wartime propaganda, so "inflicted by" (meaning caused or imposed by) is the logical fit. "Conferred on" means bestowed, which has a positive connotation incompatible with "damage."

Q5. D) To describe a recent archaeological find and its significance

The passage presents the discovery of a Roman bakery and explains how it helps scholars understand food production. It does not discuss politics (A), compare civilizations (C), or evaluate nutrition (B).

Q6. B) To identify and account for a potentially misleading tendency in the work of sociologists of industrialization

The text shows that sociologists overlook precolonial commerce in Southeast Asia, then explains why: their reliance on colonial-era archives. This identifies and accounts for a misleading tendency.

Q7. A) To comment on a human inclination that may explain why the citizens of the parish regard Mr. Bumble as they do

The narrator reflects on how people prefer simple judgments—it is easier to assume someone is beyond reproach than to examine evidence. This psychological tendency explains the citizens' uncritical view of Mr. Bumble.

Q8. B) She is eager for the gathering to conclude so that she can reclaim the space for her own purposes.

Herminia keeps checking whether guests have left because the room serves as her sewing workspace during the day, and the evening gatherings prevent her from restoring it in time.

Q9. B) The growing emphasis on environmental sustainability is a factor influencing demand for old-growth timber.

The text explicitly states that "environmental regulations increasingly mandate sustainable forestry practices," causing demand for old-growth timber to decline.

Q10. C) The gecko's thick, flexible vocal fold

The passage directly states that chirping geckos "have a thick, flexible vocal fold that vibrates when air passes over it, producing sound." This is identified as the trait enabling chirping.

Q11. B) 4.93 square kilometers.

The passage mentions Newtok's area (8.31 sq km) and asks for Kivalina's area. From the table, Kivalina = 4.93 square kilometers.

Q12. D) "Already [Gatsby's] restless mind, racing ahead to the afternoon tea..."

The claim is that Gatsby visualizes a reunion with Daisy. Option D directly shows his imagination "racing ahead" to future moments with her, illustrating this anticipatory visualization.

Q13. A) The short-snouted river dolphin collided with obstacles more frequently than the medium-snouted bottlenose dolphin.

The prediction states medium-snouted dolphins will be more successful at detecting obstacles in turbid water. If short-snouted dolphins collided more often, it confirms medium-snouted ones navigate better, directly supporting the prediction.

Q14. C) Thailand (large population, low density) compared with the Philippines (smaller area, higher density).

The student's claim is that large-population countries can be less densely populated than small-population ones. Thailand has 69.8M people at 136/km² while the Philippines has a smaller area but 368/km², illustrating this point with data from the table.

Q15. B) The right-pawedness observed in captive lions and leopards might be, in part, a consequence of prolonged exposure to human caretakers.

The passage notes captive animals spend substantially more time with humans than wild ones do. The "therefore" logically connects this observation to the conclusion that captive right-pawedness could result from human interaction.

Q16. A) cosmetic chemists,

"Developed by industrial chemists" uses a simple plural noun (chemists) as the object of "by." No possessive is needed. The comma after "chemists" sets off the following descriptive phrase ("like carriageen").

Q17. D) century, but

Two independent clauses need a conjunction: "were widely used during the... century, but these instruments were eventually replaced." A comma + coordinating conjunction correctly joins them.

Q18. C) Enwezor

The name "Okwui Enwezor" is part of a continuing sentence ("honoring curator Okwui Enwezor is exhibited..."). No punctuation should interrupt the flow between the name and the verb.

Q19. C) it's

"It's" is the contraction of "it is": the constitution "it is ranked among the longest." "Its" (possessive) is incorrect here because the sentence requires the verb "is."

Q20. D) surface, however,

"However" functions as a parenthetical interrupter within the sentence: "few particles are likely to reach Earth's surface, however, since the meteoroids' interaction..." The word "since" introduces the reason clause that follows.

Q21. B) For instance,

The preceding sentence makes a general claim (amphibians safely migrate during the day). The next sentence provides a specific mechanism of how this safety works (their color patterns deter predators), which is an illustrative example.

Q22. A) For this reason,

The metro is relied upon for millions of trips (the reason), so officials work to maintain it (the result). "For this reason" signals a cause-and-effect relationship.

Q23. A) as a result,

The preceding sentence identifies a problem (3D elements lack organic realism). The next sentence describes a consequence: post-modeling processes are needed. "As a result" correctly signals this cause-effect link.

Q24. C) Ada Spencer served as a head telegraphist in an earlier era than did Florence Nightingale Graham.

The goal is to emphasize a difference. Ada served 1874–1896; Florence served 1901–1923. Option C directly highlights this chronological difference between the two.

Q25. C) Approximately 200 million years ago, a large asteroid (Theia) may have drifted close enough to Jupiter that gravitational forces shattered it; its fragments then formed the planet's Trojan asteroids.

This option recounts the events in chronological order: Theia approached Jupiter → tidal forces broke it apart → debris formed the Trojan asteroids. The other options omit key steps or present them out of sequence.

Q26. C) While each displays distinctive features of its own, the electric blue hap and the yellow lab cichlid are both found in Lake Malawi.

The goal is to emphasize a similarity. Both species are found in Lake Malawi—a shared trait. Option C highlights this while acknowledging their individual differences.

Q27. D) Though she was known locally during her lifetime, Bouboulina's subsequent portrayal in novels, films, and television has transformed her into an internationally recognized symbol of resistance.

The goal is to emphasize the role of media. Option D directly credits media portrayals with transforming Bouboulina from a locally known figure into an international symbol—clearly attributing her expanded legacy to media.

Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Q1. A) remarkable

The passage describes an "exceptional collection" with diverse types of art. "Remarkable" (meaning notably great) captures this positive sense of impressive variety.

Q2. B) recognition

Critics "praised" the films, which constitutes recognition or acclaim. Supporters hope this positive recognition draws wider viewership.

Q3. B) trace

"Trace" means to follow or track the course of something. Anthropologists can "trace patterns" in migration—meaning follow and reconstruct those patterns.

Q4. D) It introduces a development in consumer goods and then explains how that development both advantages and poses a difficulty for companies.

The text introduces product customization, explains its benefit (consumers pay more), then identifies the challenge (hard to keep costs low with standard production methods).

Q5. D) It summarizes a favorable interpretation of a specific quality of the Riftwar novels.

The underlined portion presents the argument that the novels should be read as one continuous epic rather than separate books—a positive reframing of their unconventional structure.

Q6. B) It contrasts the frequency with which a biological phenomenon has been identified in two groups of organisms.

The sentence notes lateral gene transfer is "widespread among prokaryotes" but "seldom documented among eukaryotes"—a direct contrast of frequency between two organism groups.

Q7. A) The barriers used in Matsuda and Adeyemi's study successfully prevented root-to-root carbohydrate exchange.

Text 1 states barriers isolated root systems. Text 2 agrees they "appropriately controlled for direct tissue-to-tissue contact (thereby ruling out contact transfer)." Both authors agree on this point.

Q8. B) As ambient temperature rises, desert iguanas increase their foraging activity.

The text states temperature was "positively correlated with foraging by desert iguanas." A positive correlation means as one variable rises, so does the other.

Q9. D) Both butterfly color and weather conditions

The study found brown moths appeared more often at lights (color-related) and moths flew more on low-pressure evenings (weather-related). Both factors were linked to behavior.

Q10. D) When Vargas and colleagues played recordings of neutral ambient forest sounds near C. capucinus, the monkeys exhibited no observable response.

A control condition showing no reaction to neutral sounds would confirm that the capuchins' defensive behavior was specifically triggered by the alarm calls, not just any auditory stimulus.

Q11. C) Dr. Omar Haddad et al. found a density range greater than that found by some studies that employed other techniques.

The student claims scat-analysis dogs yield the smallest range. Haddad (scat dogs) had a range of 2.65, but Marchetti (howl-response) had 0.25—a much smaller range using a different method. This weakens the claim.

Q12. C) Petrov and colleagues' analysis revealed a slightly positive correlation between agglutination and exotericity and between tonal distinction and exotericity.

The LCH predicts negative correlation between exotericity and complexity markers. If agglutination and tonal distinction (positive complexity markers) show POSITIVE correlation with exotericity, this contradicts the LCH, supporting Petrov's claim of inconsistency.

Q13. B) Participant S12 would find landscape paintings more aesthetically pleasing than still-life paintings.

S12's model-predicted score is higher for landscape (0.28) than for still life (0.22). Therefore, the model predicted greater enjoyment for the landscape painting, so B is correct.

Q14. D) Sturluson drew on a tradition that the *skaldic* poets did not have access to when composing their works.

Since neither the skaldic poems nor earlier sources mention Mjölfnir returning to Thor's hand, but the Prose Edda does, Sturluson must have consulted a source unknown to the earlier poets.

Q15. C) some studies of fish found greater effects of noise exposure than some studies of invertebrates did.

The text states that for every taxonomic class, individual studies showed effects "well above the invertebrate average." This means some fish studies exceeded some invertebrate studies in effect size.

Q16. C) uncommon;

Two independent clauses: "these consonants are uncommon" and "in several Caucasian languages, however, five or more... are regularly used." A semicolon correctly joins them.

Q17. B) germination:

A colon introduces the explanatory elaboration: "sunlight exposure influences the growth rate of germination: as sunlight exposure increases, so does the rate..." The colon signals that what follows explains the preceding claim.

Q18. C) develop

The subject is "such symbiotic pairings" (plural), requiring the plural present-tense verb "develop." Infinitive ("to develop") and participial forms ("developing") would create fragments.

Q19. D) conductor, Tõnu Kaljuste,

"Tõnu Kaljuste" is an appositive renaming "conductor." Appositives are set off by commas on both sides: "Estonian conductor, Tõnu Kaljuste, may be best celebrated..."

Q20. B) Goldstein; however,

Two independent clauses: "The anthology was not composed solely by Goldstein" and "other marine scientists... also contributed chapters." A semicolon + "however," correctly joins them.

Q21. D) nation—Poland.

An em dash introduces a specific identification of the unnamed "fellow member nation." This is standard usage for appositional dashes that reveal or specify.

Q22. B) That said,

The preceding sentences state a limitation (Lin cannot literally depict a fifth dimension). "That said" (meaning "nevertheless" or "even so") introduces a contrasting point: she still creates an effective impression.

Q23. C) Therefore,

Nakamura argued the Society's calculation overlooked critical variables (the reason). "Therefore" introduces the logical consequence: the Society's designated center was inaccurate.

Q24. D) Indeed,

The previous sentence notes a "significant number" of receptions honored aviation achievements. "Indeed" reinforces and specifies: Yeager's was just one of 22 such events.

Q25. B) Josef Albers and Anni Albers were both Bauhaus students, but Josef, who enrolled in 1920, preceded Anni by two years.

This directly states both were students and highlights the chronological order of their enrollment (Josef in 1920, Anni in 1922).

Q26. A) The copper-nickel alloy sample tested in the 2020 study was thicker than the dimethicone sample.

The goal is to contrast the two samples. Option A notes a key difference: the copper-nickel sample was 5mm thick versus the dimethicone sample's 1mm thickness.

Q27. A) The mass fraction of quartz in Earth's continental crust is 120,000 ppm.

The goal is to specify quartz's mass relative to all minerals. The mass fraction (120,000 ppm) is precisely that ratio, expressed as a concrete figure. Option C describes what a mass fraction is but doesn't provide the specific value.

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Q1. A) expansive

The passage describes diversity from potatoes to lesser-known oca across different altitudes, so "expansive" (meaning broad or wide-ranging) best fits the context of extensive agricultural variety.

Q2. D) endorsement

Critics "lauded" the works, which constitutes endorsement (strong public approval). Supporters hope this positive endorsement will attract new readers to the genre.

Q3. B) reconstruct

"Reconstruct" means to piece together or rebuild an understanding of something. Prehistorians can "reconstruct transitions" in early societies—fitting the scholarly context of interpreting fragmentary evidence.

Q4. A) To reflect on a psychological tendency that may account for why Jim feels attached to the prairie landscape

The narrator observes that "the familiar is always simpler to cherish" and requires "no argument, no comparison." This is a reflection on a general psychological tendency that explains Jim's attachment.

Q5. A) It draws a contrast between the prevalence of a biological process in two categories of organisms.

The sentence contrasts epigenetic modification being "well documented in multicellular organisms" with it being "long thought to be absent in unicellular organisms"—a direct frequency contrast.

Q6. A) The enclosures used in Okafor and Halvorsen's study effectively prevented tissue-to-tissue compound transfer.

Text 1 states they used enclosures to isolate colonies. Text 2 agrees they "appropriately controlled for direct tissue-to-tissue contact." Both authors accept that this control was effective.

Q7. B) In pastures where artificial lighting eliminated the distinction between day and night, one of the two species declined markedly within three years.

If removing the day-night distinction (the basis for niche partitioning) causes competitive exclusion, this directly supports the hypothesis that temporal differences in activity patterns are what allow coexistence.

Q8. B) As the volume of air traffic increased, experienced controllers maintained a stable error rate.

The text states Ishikawa found "no significant increase in errors when managing high-traffic versus low-traffic airspace." A stable error rate despite increased complexity is the finding.

Q9. A) Region X would produce more grain crops than legume crops.

For Region X, the model-predicted yield index is higher for grain (0.30) than for legumes (0.18). With equal acreage, the model therefore predicts greater grain output, so A is correct.

Q10. D) Colombe drew on an iconographic convention that the Limbourg Brothers had not employed when creating their portion of the manuscript.

Since neither the Limbourg Brothers' original sections nor any earlier illuminated works use gold leaf halos for the patron, but Colombe's additions do, he must have drawn on a tradition unknown to or unused by his predecessors.

Q11. C) some studies of fish found greater effects of noise exposure than some studies of invertebrates did.

The text states that for every taxonomic class, "individual studies showing differences well above the invertebrate average" existed. Thus some fish studies exceeded some invertebrate studies in measured effect.

Q12. D) When Engström and colleagues dispersed a neutral, odorless control substance near *F. rufa* colonies, the ants showed no behavioral change.

A null result with a control substance would confirm that the ants' defensive behavior was specifically triggered by the alarm chemicals rather than by any general disturbance.

Q13. C) absent;

Two independent clauses joined by a semicolon: "these consonants are absent; in several Semitic languages, however, three or more such sounds are regularly produced."

Q14. B) water:

The colon introduces an explanatory elaboration of how barometric pressure affects evaporation rate: "...evaporation rate of water: as barometric pressure decreases, so does the boiling point..."

Q15. C) produce

The subject is "such symbiotic relationships" (plural), requiring the plural present-tense verb "produce."

Q16. D) dancer, Marcia Haydée,

"Marcia Haydée" is an appositive renaming "dancer" and must be set off by commas on both sides.

Q17. B) Johansson; however,

Two independent clauses joined by semicolon + conjunctive adverb: "The compendium was not written solely by Johansson; however, other researchers... also contributed."

Q18. D) state—Mexico.

The em dash introduces the specific identification of the member state, a standard appositional dash construction.

Q19. B) Even so,

The preceding text states a limitation (Klimt's artwork cannot literally render a fourth dimension). "Even so" (meaning "despite that") transitions to the contrasting point that he still creates a compelling impression.

Q20. C) Consequently,

Osei argued the Institut neglected important factors (cause). "Consequently" introduces the logical result: the established center was inaccurate.

Q21. C) In fact,

The previous sentence notes a "notable proportion" of state dinners honored aviation. "In fact" intensifies this by providing the specific figure: just 9 honored solo aviators.

Q22. B) Both Hughes and Hurston were active participants in the Harlem Renaissance, a cultural movement centered in New York City.

The goal is to emphasize a similarity. Option B highlights what they share: both were active participants in the same movement.

Q23. C) The age of the Jack Hills zircon can be expressed by a radiometric date, a measure derived from the ratio of remaining radioactive isotopes in the specimen.

The goal is to specify age "relative to that of all known mineral specimens." Option C explains that a radiometric date measures isotope ratios, which IS a relative measure—addressing "relative to all" conceptually.

Q24. A) The krypton sample in the 2021 study was cooled through a larger temperature range than the argon sample and consequently contracted more.

This directly contrasts the two samples: krypton was cooled 100 K (300→200) versus argon's 50 K (300→250), and krypton contracted more (166.7 mL vs 83.3 mL).

Q25. C) When the gamble was reframed as avoiding a \$100 penalty rather than risking a \$100 loss, acceptance rates increased significantly.

If merely changing the description from "loss" to "avoidance" increases acceptance—despite identical mathematical outcomes—this demonstrates that the psychological frame of "loss" specifically drives the aversion, supporting loss aversion theory.

Q26. C) Site B, which used camera traps, produced a density range larger than that found by some studies that used other techniques.

While the hypothesis about acoustic monitoring is supported by the data (it does have the smallest range), option C highlights that methods vary in precision, providing context for evaluating the claim.

Q27. A) linguistic structure influences certain categories of spatial cognition without uniformly determining all aspects of thought.

Tzeltal speakers outperformed on spatial rotation but matched Dutch speakers on object recognition. This partial effect contradicts the strong Sapir-Whorf hypothesis (which predicts uniform determination) while supporting a weaker version.

Section 2: Math

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Q1. B) -2

The two lines intersect at the point (3, -2). Since the solution is (x, y), the value of y is -2.

Q2. C) 43

$$g(3) = 12(3) + 7 = 36 + 7 = 43.$$

Q3. D) $f(x) = 5x + 17$

The line passes through (0, 17), so the y-intercept is 17. The slope is 5. Using slope-intercept form: $f(x) = 5x + 17$.

Q4. A) $y = x + 30$

When $x = 0$, $y = 30$; when $x = 1$, $y = 31$; when $x = 2$, $y = 32$. The slope is 1 and the y-intercept is 30, giving $y = x + 30$.

Q5. D) 4

For infinitely many solutions, both equations must be identical. The second equation $y = mx + b$ must equal $y = (3/7)x + 4$, so $b = 4$.

Q6. C) \$3.00

$$25\% \text{ of } \$4.00 = \$1.00 \text{ discount. Sale price} = \$4.00 - \$1.00 = \$3.00.$$

Q7. A) $y = 75(1.15)^x$

Initial value = 75 (when $x = 0$). Growth rate = 15% per unit increase, so the multiplier is 1.15. Equation: $y = 75(1.15)^x$.

Q8. B) 12

$$\text{Mean} = (10 + 8 + 16 + 14 + 12) \div 5 = 60 \div 5 = 12.$$

Q9. D) 27

$$h(-3) = 9|-3| = 9 \times 3 = 27. \text{ The absolute value makes the input positive before multiplying.}$$

Q10. B) -1.12

The line of best fit goes from approximately (0, 14) to (8, 5.5). Slope $\approx (5.5 - 14)/(8 - 0) = -8.5/8 \approx -1.06$. The closest option is -1.12.

Q11. A) 25

With parallel lines cut by a transversal, alternate interior angles are equal. The angle at line p (80°) equals y° , so $y = 80$. Then $80 = 3x + 5$, giving $3x = 75$ and $x = 25$.

Q12. Grid-in: 18

$$\text{Let } p = \text{price per scarf. } 7p - 35 = 91. 7p = 126. p = 18.$$

Q13. C) $y = -x/4 + 7$

$$\text{Using point-slope form: } y - 5 = -(1/4)(x - 8). y = -x/4 + 2 + 5 = -x/4 + 7.$$

Q14. Grid-in: 9 or 14

Setting each factor to zero: $x = 14$, 9, -5, or -11. The positive solutions are 9 and 14. Either is acceptable.

Q15. C) 80π

$$\text{Volume} = (1/3)\pi r^2 h = (1/3)\pi(4^2)(15) = (1/3)\pi(240) = 80\pi.$$

Q16. Grid-in: 21000

Linear model: from 2000 to 2012 (12 years), enrollment rose from 8,000 to 14,500 (increase of 6,500). Rate = 6,500/12 per year. From 2012 to 2024 (another 12 years): increase = 6,500. $x = 14,500 + 6,500 = 21,000$.

Q17. A) The estimated number of subscribers at the end of March 2020 was 400.

The y-intercept occurs when $x = 0$. $y = 400(3)^0 = 400(1) = 400$. Since $x = 0$ corresponds to the end of March 2020 (the starting point), the answer is 400 subscribers at that time.

Q18. Grid-in: 2800

Adding the two equations eliminates y : $x/5 + x/4 = 47 + 16 = 63$. Thus $9x/20 = 63$, so $x = 140$ and $20x = 2,800$.

Q19. B) $0 < x < 80$

Total capacitance = $200 \mu\text{F}$. Four capacitors sum to $120 \mu\text{F}$. Remaining six sum to $80 \mu\text{F}$. Each of the six is positive, so for any one capacitor x : $0 < x < 80$ (since the other five must also be positive).

Q20. B) 2

In $f(n) = 25 - 2n$, the coefficient -2 means the balance decreases by \$2 for each book borrowed. Therefore, the amount deducted each time is 2 dollars.

Q21. Grid-in: -1

Let $u = 5 - 2x$. Then $7 - 3u = 4 - 6u$. Solving: $-3u + 6u = 4 - 7 \rightarrow 3u = -3 \rightarrow u = -1$. So $5 - 2x = -1$.

Q22. Grid-in: 15.12

Barns on grazing land: $18.5\% \times 48.0\% = 8.88\%$. Barns on cropland: $12.0\% \times 52.0\% = 6.24\%$. Total: $p = 8.88 + 6.24 = 15.12$.

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Q1. D) 62

In a right triangle, angles sum to 180° . $90 + 28 + a = 180$, so $a = 62$.

Q2. A) 2

If the graph passes through $(7, 2)$, then $f(7) = 2$ by definition of the function.

Q3. Grid-in: 56

Circumference $= \pi d$. $56\pi = \pi d$, so $d = 56$.

Q4. A) $v = \sqrt{(2K/36)}$

Starting from $36 = 2K/v^2$, multiply both sides by v^2 : $36v^2 = 2K$. Divide by 36: $v^2 = 2K/36$. Take the square root: $v = \sqrt{(2K/36)}$.

Q5. D) 300

$f(10) = 3(10)^2 = 3(100) = 300$.

Q6. B) $(0, 1/7)$

The y-intercept is at $x = 0$: $g(0) = (1/7)(4^0) = (1/7)(1) = 1/7$. The point is $(0, 1/7)$.

Q7. D) 6

3 cups flour : 2 cups sugar. With 9 cups flour ($\times 3$), sugar $= 2 \times 3 = 6$ cups.

Q8. Grid-in: 1000

Population tripled every 60 years over 240 years (1700-1940), or 4 periods. $81,000 = P_0 \cdot 3^4 = 81P_0$, so $P_0 = 1,000$.

Q9. C) 36

With $\tan(R) = PQ/QR = 3/5$ and interpreting the labeled side as adjacent to R: if $QR = 60$, then $PQ = (3/5)(60) = 36$.

Q10. C) $y = 5(x - 1) + 8$

First movie costs \$8. Each additional movie costs \$5. For x movies: $y = 8 + 5(x - 1) = 5(x - 1) + 8$.

Q11. B) -7 only

Expand both sides: $a(5x - 10) + 3a = 5ax - 7a$ and $5(ax - 2a) - 21 = 5ax - 10a - 21$. For infinitely many solutions, the constants must match: $-7a = -10a - 21$, so $a = -7$. Therefore, B is correct.

Q12. A) $(100/25)x$

If $x = 0.25y$, then $y = x/0.25 = 4x = (100/25)x$.

Q13. Grid-in: 4/3

$x^{m/4} = x^{1/3}$. So $m/4 = 1/3$, giving $m = 4/3$.

Q14. C) 44.8

$z(w+1)/z(w) = (0.743)^2 = 0.5521$. The function retains 55.21% and decreases by 44.79%. So $p \approx 44.8$.

Q15. A) $f(x) = -d - cx$

The graph of $y = f(x) + 9$ has a positive y-intercept and steep negative slope. At $x = 0$: $f(0) + 9 \approx 6$, so $f(0) \approx -3$. With positive c and d : $f(x) = -d - cx$ gives $f(0) = -d$ (negative) and negative slope $-c$. This matches.

Q16. D) $(24 \times 63 \times k)/m$

Rate: 63 widgets per m hours $= 63/m$ per hour. Per day: $63 \times 24/m$. Per k days: $63 \times 24 \times k/m = (24 \times 63 \times k)/m$.

Q17. Grid-in: 1/15

By Vieta's formulas, the product of solutions $= rs/15$. Since the product equals kr : $k = 1/15$.

Q18. A) 16,717c

$a = 50c$ (5,000% of c). $b = c/0.00006 = 50,000c/3$. $a + b = 50c + 50,000c/3 = (150c + 50,000c)/3 = 50,150c/3 \approx 16,716.7c \approx 16,717c$.

Q19. C) 44

Expanding gives $-36x^2 + 186x - 139$. Matching $(a/4)x^2 + (b/4)x + c/4$ gives $a = -144$, $b = 744$, and $c = -556$. Hence $a + b + c = 44$, so C is correct.

Q20. D) $100(b - 1)$

$f(x+1)/f(x) = b$. The percentage increase is $(b - 1) \times 100 = c\%$. So $c = 100(b - 1)$.

Q21. D) $(4 \times 52 \times k)/m$

Rate: 52 cm per m weeks = $52/m$ per week. Per month (4 weeks): $52 \times 4/m$. Per k months: $52 \times 4 \times k/m = (4 \times 52 \times k)/m$.

Q22. Grid-in: 40

PR is a diameter of length 205 and is perpendicular to chord QT, so it bisects QT. Thus $QS = ST = 20$. By the intersecting-chords theorem, $PS(RS) = 20(20) = 400$ and $PS + RS = 205$. The segment lengths are 5 and 200. Since $PQ < QR$, PS is the shorter segment, so $r = RS/PS = 200/5 = 40$.

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Q1. B) 71

In a right triangle, angles sum to 180° . $90 + 19 + b = 180$, so $b = 71$.

Q2. D) 6

If the graph passes through $(4, 6)$, then $f(4) = 6$.

Q3. Grid-in: 38

Circumference $= \pi d$. $38\pi = \pi d$, so $d = 38$.

Q4. A) $v = \sqrt{(2K/64)}$

From $64 = 2K/v^2$: $v^2 = 2K/64$, so $v = \sqrt{(2K/64)}$.

Q5. D) 405

$f(9) = 5(9)^2 = 5(81) = 405$.

Q6. B) $(0, 1/11)$

$g(0) = (1/11)(7^0) = (1/11)(1) = 1/11$. The y-intercept is $(0, 1/11)$.

Q7. Grid-in: 1260

Adding the equations: $x/6 + x/2 = 70 \rightarrow (x + 3x)/6 = 70 \rightarrow 4x/6 = 70 \rightarrow x = 105$. So $12x = 1,260$.

Q8. Grid-in: 100

Bacteria quadrupled every 50 minutes over 150 minutes (3 periods). $6,400 = P_0 \cdot 4^3 = 64P_0$, so $P_0 = 100$.

Q9. C) 60

With $\tan(F) = DE/EF = 5/7$ and interpreting the labeled side as adjacent: if $EF = 84$, then $DE = (5/7)(84) = 60$.

Q10. B) -9 only

Expanding: $9ax - 18a + 7a = 9ax - 18a - 63 \rightarrow -11a = -18a - 63 \rightarrow 7a = -63 \rightarrow a = -9$.

Q11. A) $(100/40)x$

If $x = 0.40y$, then $y = x/0.40 = 2.5x = (100/40)x$.

Q12. Grid-in: 5/3

$x^{m/5} = x^{1/3}$. So $m/5 = 1/3$, giving $m = 5/3$.

Q13. C) $y = 9(x - 1) + 15$

First album: \$15. Each additional: \$9. For x albums: $y = 15 + 9(x - 1) = 9(x - 1) + 15$.

Q14. D) 15.9

$z(w+1)/z(w) = (0.917)^2 = 0.8409$. The function retains 84.09% and decreases by 15.91%. So $p \approx 15.9$.

Q15. A) $f(x) = -d - cx$

The graph of $y = f(x) + 7$ has a positive y-intercept and negative slope. $f(0) + 7 \approx 4$ means $f(0) \approx -3$. With c, d positive: $f(x) = -d - cx$ gives $f(0) = -d$ (negative) and slope $-c$ (negative). This matches.

Q16. D) $100(b - 1)$

$f(x+1)/f(x) = b$. The increase is $(b - 1) \times 100 = c\%$. So $c = 100(b - 1)$.

Q17. Grid-in: 1/19

By Vieta's formulas, the product of solutions $= rs/19$. Since the product equals kr : $k = 1/19$.

Q18. A) 20,040c

$a = 40c$ (4,000% of c). $b = c/0.00005 = 20,000c$. $a + b = 40c + 20,000c = 20,040c$.

Q19. C) -36

Expanding gives $-144x^2 + 408x - 273$. Matching $(a/4)x^2 + (b/4)x + c/4$ gives $a = -576$, $b = 1,632$, and $c = -1,092$. Therefore, $a + b + c = -36$, so C is correct.

Q20. D) $(60 \times 48 \times k)/m$

Rate: 48 pages per m minutes = $48/m$ per minute. Per hour (60 min): $48 \times 60/m$. Per k hours: $(60 \times 48 \times k)/m$.

Q21. B) $0 < w < 120$

Total = 300 W. Five bulbs = 180 W. Remaining seven = 120 W. Each is positive, so for one bulb w : $0 < w < 120$.

Q22. Grid-in: 25

WY is a diameter of length 260 and is perpendicular to chord XZ, so it bisects XZ. Thus $XV = VZ = 50$. By the intersecting-chords theorem, $WV(VY) = 2,500$ and $WV + VY = 260$. The segment lengths are 10 and 250. Since $WX < XY$, WV is shorter, so $r = YV/WV = 250/10 = 25$.

Quick-Reference Answer Grid

Section 1, Module 1

Q#	Answer
1	A
2	A
3	C
4	B
5	B
6	B
7	A
8	B
9	B
10	C
11	B
12	D
13	A
14	C
15	B
16	A
17	D
18	C
19	C
20	B
21	B
22	A
23	A
24	C
25	C
26	C
27	D

Section 1, Module 2 — Easier

Q#	Answer
1	A
2	B
3	B
4	D
5	D
6	B
7	A
8	B
9	D
10	D
11	C
12	C
13	B
14	D
15	C
16	C
17	B
18	C
19	D
20	B
21	D
22	B
23	C
24	C
25	B
26	A
27	A

Section 1, Module 2 — Harder

Q#	Answer
1	A
2	B
3	B
4	A
5	A
6	A

7	B
8	B
9	B
10	D
11	C
12	D
13	C
14	B
15	C
16	D
17	B
18	D
19	B
20	C
21	C
22	B
23	C
24	A
25	C
26	C
27	A

Section 2, Module 1

Q#	Answer
1	B
2	C
3	D
4	A
5	D
6	C
7	A
8	B
9	D
10	B
11	A
12	18
13	C
14	9 or 14
15	C
16	21000
17	B
18	2800
19	B
20	B
21	-1
22	15.12

Section 2, Module 2 — Easier

Q#	Answer
1	B
2	A
3	56
4	A
5	D
6	B
7	C
8	1000
9	C
10	C
11	B

12	A
13	4/3
14	C
15	A
16	D
17	1/15
18	A
19	C
20	D
21	D
22	40

Section 2, Module 2 — Harder

Q#	Answer
1	B
2	B
3	38
4	A
5	D
6	B
7	1260
8	100
9	C
10	B
11	A
12	5/3
13	C
14	C
15	A
16	D
17	1/19
18	A
19	C
20	D
21	B
22	25

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